

Final Project Proposal

Nominating Drugs and Their Alternatives Based on Active Ingredient and Side Effects

1. Project Description

This project aims to develop an intelligent drug recommendation system that assists healthcare professionals and patients in identifying suitable drug alternatives based on active ingredients and side effect profiles. Using Retrieval-Augmented Generation (RAG) technology, the system will:

- Accept user queries about specific drugs or medical conditions
- Retrieve relevant information from a comprehensive pharmaceutical database
- Analyze active ingredients and associated side effects
- Generate personalized drug recommendations and alternatives
- Provide detailed comparisons between drugs with similar therapeutic effects but different side effect profiles

The system will leverage natural language processing to understand user queries and provide accurate, context-aware recommendations while maintaining a focus on drug safety and efficacy. This solution addresses the critical need for accessible drug information and alternative suggestions in healthcare decision-making.

2. Group Members & Roles

Name	Role	Responsibilities
Abdullah Muhamed Abuslama	Team Leader & MLOps Engineer	Project coordination, pipeline integration, deployment strategy, version control, team communication, and ensuring milestone completion
Ibrahim Rashad Ali Waz	Data Engineer & RAG Specialist	Data collection and preprocessing, database design, RAG architecture implementation, vector store management, and retrieval optimization
Ahmed Mohamed Ali ElBadry	ML Engineer & Model Developer	Model selection and fine-tuning, prompt engineering, LLM integration, performance optimization, and evaluation metrics implementation
Mohamed Hossam Hassan Nabil	Backend Developer & API Designer	API development, system integration, user interface development, testing, and documentation

3. Team Leader

Abdullah Muhamed Abuslama

4. Project Objectives

Primary Objectives:

1. **Build an Accurate Drug Recommendation System:** Develop a RAG-based system that accurately retrieves and recommends drug alternatives based on active ingredients with a minimum accuracy of 85%
2. **Comprehensive Side Effect Analysis:** Create a robust database and analysis module that compares side effect profiles across different drugs to recommend safer alternatives
3. **User-Friendly Query System:** Implement a natural language interface that allows users to query drugs using simple medical terminology or drug names
4. **Real-Time Information Retrieval:** Ensure the system retrieves relevant drug information in under 3 seconds per query
5. **Scalable and Maintainable Solution:** Design a production-ready system with proper MLOps practices, version control, and deployment pipelines

Secondary Objectives:

- Provide detailed drug information including dosage, contraindications, and interactions
 - Support multiple query formats (by drug name, active ingredient, or medical condition)
 - Generate comprehensive comparison reports between alternative drugs
 - Ensure system reliability with 95%+ uptime in deployment environment
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5. Tools & Technologies

Core Technologies:

- **Programming Language:** Python 3.10+
- **RAG Framework:** LangChain / LlamaIndex
- **Large Language Model:** OpenAI GPT-4 / Anthropic Claude / Open-source alternatives (Llama 3, Mistral)
- **Vector Database:** Pinecone / ChromaDB / Weaviate
- **Embeddings:** OpenAI Embeddings / Sentence-Transformers

Data & Database:

- **Data Sources:** FDA Drug Database, DrugBank, RxNorm, Medical literature
- **Database:** PostgreSQL / MongoDB for structured data
- **Data Processing:** Pandas, NumPy

MLOps & Deployment:

- **Experiment Tracking:** MLflow / Weights & Biases

- **Version Control:** Git & GitHub
- **Containerization:** Docker
- **API Framework:** FastAPI / Flask
- **Cloud Platform:** AWS / Google Cloud / Azure (for deployment)

Development & Testing:

- **IDE:** VS Code / PyCharm
- **Notebooks:** Jupyter Notebook / Google Colab
- **Testing:** Pytest, Postman
- **Documentation:** Sphinx / MkDocs

6. Milestones & Deadlines

Milestone	Description	Deadline	Deliverables
M1: Data Collection & Preparation	Gather drug datasets, preprocess data, create vector embeddings	End of Week 16	- Cleaned dataset - Data preprocessing scripts - Initial vector database
M2: RAG System Development	Implement RAG architecture, vector store setup, retrieval mechanism	Mid Week 17	- Functional RAG pipeline - Retrieval testing results - Basic query interface
M3: Model Integration & Testing	Integrate LLM, prompt engineering, response generation	End of Week 17	- Complete RAG system - Prompt templates - Test results document
M4: Prototype Development	API development, basic UI, system integration	Mid Week 18	- Working Prototype - API documentation - Demo interface
M5: Evaluation & Optimization	Performance testing, accuracy evaluation, system optimization	End of Week 18	- Evaluation report - Performance metrics - Optimization log
M6: MLOps & Deployment	Implement CI/CD, containerization, deployment setup	Mid Week 19	- Deployment pipeline - Docker containers - Monitoring setup
M7: Documentation & Finalization	Complete documentation, prepare presentation, final testing	End of Week 19	- Complete documentation - Presentation slides - User guide
M8: Final Presentation	Project demonstration and presentation	Week 20	- Final Presentation - Live demo - Q&A session

7. Key Performance Indicators (KPIs)

1. Data Quality & Coverage Score (Target: ≥90%)

Measures: The completeness, accuracy, and coverage of the pharmaceutical database

- **Metrics:**

- Percentage of drugs with complete information (active ingredients, side effects, contraindications)
- Data accuracy validation score
- Coverage of common medications (target: 1000+ drugs)
- Embedding quality score

2. Retrieval Accuracy & Relevance (Target: $\geq 85\%$)

Measures: How accurately the RAG system retrieves relevant drug information for queries

- **Metrics:**
 - Retrieval precision and recall
 - Top-K accuracy (relevant documents in top 5 results)
 - Semantic similarity scores
 - Context relevance evaluation

3. Recommendation Accuracy & Safety (Target: $\geq 90\%$)

Measures: Correctness of drug recommendations and alternatives suggested by the system

- **Metrics:**
 - Recommendation accuracy (validated against medical guidelines)
 - Active ingredient matching accuracy
 - Side effect profile comparison accuracy
 - False positive rate for contraindications (target: $< 5\%$)

4. System Performance & Response Time (Target: < 3 seconds)

Measures: Speed and efficiency of the complete pipeline from query to response

- **Metrics:**
 - Average query response time
 - Vector search latency
 - LLM generation time
 - System throughput (queries per minute)

5. Output Quality & Clinical Usefulness (Target: $\geq 4/5$)

Measures: Quality, clarity, and medical usefulness of generated recommendations

- **Metrics:**
 - Expert evaluation score (by medical professionals or instructors)
 - Information completeness score

- Clarity and readability rating
- Actionability of recommendations

6. MLOps Maturity & System Reliability (Target: Level 3/4)

Measures: Implementation of MLOps best practices and system maintainability

- Metrics:
 - Version control completeness (code, data, models)
 - Automated testing coverage (target: >70%)
 - Pipeline automation level
 - System uptime during demo period (target: 95%)
 - Model monitoring and logging implementation

7. Documentation & Knowledge Transfer Quality (Target: ≥90%)

Measures: Completeness and clarity of all project documentation

- Metrics:
 - Code documentation coverage
 - API documentation completeness
 - User guide clarity rating
 - Presentation effectiveness score
 - Reproducibility score (can another team run the system?)

8. Risk Assessment & Mitigation

Risk	Impact	Mitigation Strategy
Data quality issues	High	Multiple data sources, validation scripts, expert review
LLM API costs/limits	Medium	Use open-source alternatives, implement caching
Retrieval accuracy	High	Extensive testing, fine-tune embedding models, use hybrid search
Timeline delays	Medium	Weekly checkpoints, parallel task execution, buffer time
Integration challenges	Medium	Early integration testing, modular design, clear interfaces

9. Success Criteria

The project will be considered successful if:

- ✓ The system accurately retrieves drug information with ≥85% precision
- ✓ Recommendations are validated as medically sound by evaluators

3.  Response time is consistently under 3 seconds
 4.  All milestones are completed by their deadlines
 5.  Working prototype is demonstrated in Week 18
 6.  Final system is deployed and documented by Week 20
 7.  All KPIs meet or exceed target thresholds
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10. Conclusion

This project represents a practical application of RAG technology in the healthcare domain, addressing a real-world need for accessible drug information and alternative recommendations. Our team is committed to delivering a high-quality, production-ready system that demonstrates both technical excellence and practical utility.

We look forward to bringing this vision to life and presenting a solution that can make a meaningful impact in healthcare decision support.

Proposal Submitted By:

Abdullah Muhamed Abuslama (Team Leader)

On behalf of the project team

Date: October 12, 2025